

Find what matters: the best upgrade is not the asset that looks strongest; it is the intervention whose benefit survives uncertainty.

Question: With budget for one upgrade, should the utility reinforce the existing B-C bottleneck or build a new alternate path out of Region C -- and does that choice remain reliable across Monte Carlo capacity uncertainty?

The tested grid has 300 buses, 422 lines, and only one 1,800 MW B-C tie, while Region C has 20,667 MW generation and A+B have 20,999 MW load.

The **first** question is: what if the utility simply strengthens the known B-C bottleneck, BUS0161-BUS0252? That line already looks important because it connects the generation-heavy Region C to the rest of the system. But looking at it is not enough. I only treat the upgrade as a real improvement if the 95% confidence interval stays below zero for the change in weighted average path length. In plain terms, the model must show that the upgrade consistently makes the network structurally easier to move through, not just better in one lucky simulation run.

The **second** question is: what if the utility uses the same one-upgrade budget to add a new alternate path out of Region C instead? This is the main point of decision. Reinforcing BUS0161-BUS0252 makes the existing bridge stronger, while adding an alternate intertie gives Region C another route into the load-heavy side of the grid. So, the comparison is really between more capacity on the same dependency and new redundancy around that dependency. The better option is not just the one with the biggest average improvement; it is the one whose improvement is reliable enough to trust and large enough to matter.

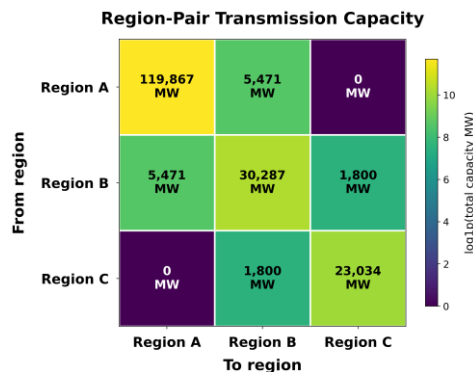
The **last** question is whether the result holds up as the Monte Carlo simulation gets larger. A small run can give a result that looks convincing but is just noise. To check this, I compare $R = 100$, $R = 500$, and $R = 2,000$. The $R = 2,000$ case is used as the benchmark. If the confidence intervals shrink in a predictable way, and a smaller run gets within 10% of the $R = 2,000$ confidence interval width, then the decision is more stable. This matters because the recommended upgrade should not depend on one favorable set of random capacity draws.

This report asks whether the best grid upgrade is to make the known bottleneck stronger or to give the system another way around it. I compare those interventions using counterfactual Monte Carlo simulation: line capacities are redrawn across plausible grid states, each intervention is applied, and the change in weighted average path length is measured against the baseline. I treat the result as actionable only if the 95% confidence interval stays on the beneficial side of zero and the mean improvement clears a pre-declared practical threshold.

Network Context:

The model represents the fictional transmission grid as an undirected weighted graph. Buses are nodes, transmission lines are edges, and line capacity in MW defines structural distance through the cost transformation cost = $1/\text{capacity}$. The baseline grid contains 300 buses and 422 transmission lines. The network is connected, but sparse: a connected grid can still depend heavily on a small number of bridges.

The regional imbalance drives the counterfactual question. Regions A and B are load-heavy, with 20,999 MW of load and 5,352 MW of generation. Region C is generation-heavy, with 20,667 MW of generation and only 254 MW of load. Structurally, Region C must export through limited inter-region pathways.



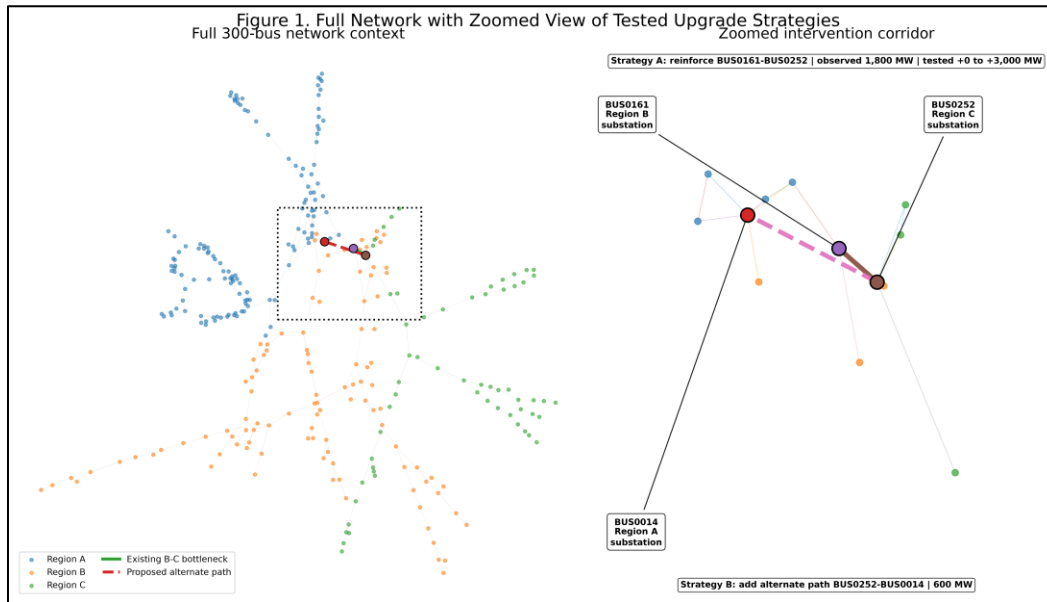
The region-pair capacity numbers explain why the B-C intertie matters. Region A has 119,867 MW of internal A-A line capacity. Region B has 30,287 MW internal B-B capacity, and Region C has 23,034 MW internal C-C capacity. The inter-region links are much thinner: A-B totals 5,471 MW, B-C is a single 1,800 MW tie, and A-C has 0 MW. That makes the B-C connection important even though it is not the largest asset by local capacity alone.

The known bottleneck from the first analysis remains BUS0161-BUS0252. BUS0161 is a Region B substation with degree 4 and 4,099 MW of incident line capacity. In the normalized summary used for this final report, BUS0161 is the top capacity-weight betweenness bus. BUS0252 is a Region C substation and the shared endpoint of the tested alternate path. BUS0014 is a high-betweenness Region A substation used as the alternate-path endpoint in this counterfactual design.

Network Intervention Context Table:

Option	Intervention	What it tests	Interpretation risk
Baseline	No upgrade after each capacity redraws.	Defines the paired comparison reference case.	Must use the same Monte Carlo draw as each intervention.
B-C capacity sweep	Add 0 to 3,000 MW to BUS0161-BUS0252.	Tests on whether strengthening the known bottleneck reduces weighted APL.	Does not create a new route around the bottleneck.
Alternate path	Add 600 MW between BUS0252 and BUS0014.	Tests whether redundancy out of Region C beats reinforcing the same corridor.	Result depends on the chosen endpoint and MW rating.

Figure 1: Pictorial Network Intervention Diagram: Shows the existing B-C bottleneck, BUS0161-BUS0252, and the tested alternate path, BUS0252-BUS0014. Region C contains 20,667 MW of generation and only 254 MW of load, while Regions A+B contain 20,999 MW of load. The existing B-C intertie has 1,800 MW capacity, so the tested decision compares adding capacity to the same corridor against creating a second route out of Region C.



State $N = 300$, $E = 422$, Region C generation = 20,667 MW, A+B load = 20,999 MW, B-C capacity = 1,800 MW, and BUS0161 as the top weighted-betweenness bus in this normalized output.

Counterfactuals Tested

The report tests actions, not just assets. The first report established that capacity is not redundancy. This report asks whether a proposed change improves the system under uncertainty.

- Baseline: no upgrade after the Monte Carlo capacity redraws. The baseline uses the same capacity draw as each intervention, which makes the comparison paired and fair.
- B-C reinforcement: add capacity to the existing BUS0161-BUS0252 corridor. Rather than assuming one upgrade size, the final optimizer tests 0, 250, 500, 750, 1,500, 2,000, and 3,000 MW additions.
- Alternate path: add a 600 MW path between BUS0252 and BUS0014. This is a specified counterfactual, not a search across every possible new line endpoint.

Critical review argument: if the alternate path wins, the report should not claim that any new line is automatically better than reinforcement. It can only claim that the tested BUS0252-BUS0014 path, at the tested 600 MW capacity and under the selected uncertainty model, performs better on the selected graph metric.

Method

Each Monte Carlo replicate redraws every observed line capacity from a Poisson distribution centered on the observed capacity. That creates a plausible perturbed version of the same grid rather than treating the observed capacity table as exact. This is a modeling assumption around observed line capacity, not a claim that thermal ratings physically fluctuate as Poisson random variables.

The simulation is paired. In each replicate, I compute the baseline weighted average path length and then apply the interventions to the same redrawn grid. The effect is intervention minus baseline. Negative values are beneficial because they mean the upgrade reduced weighted average path length.

The final run uses $R = 2,000$ Monte Carlo replicates. The baseline weighted average path length across replicates is approximately 0.036404. I also track stranded load MW as a secondary continuity metric, but in this additive-intervention design it is expected to remain zero because the experiment adds capacity or adds a line rather than removing equipment.

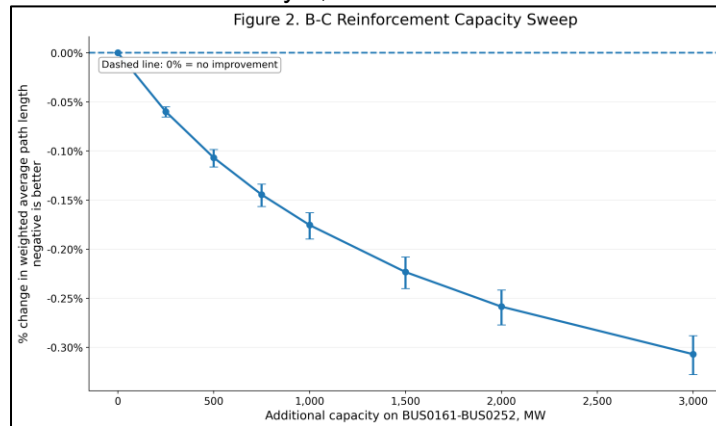
Decision criteria were stated before interpreting results. First, an intervention is statistically real only if the 95% confidence interval for intervention minus baseline stays below zero. Second, an intervention is practically meaningful only if it clears the 2% reduction threshold in weighted average path length. These two criteria intentionally separate “detectable” from “large enough to act on.”

Results

Result 1: B-C reinforcement is statistically real, but the effect is small

The smallest tested B-C upgrade with a confidence interval below zero is BC_plus_250MW, with a mean percent change of -0.060% and a 95% CI of [-0.065%, -0.055%]. This means even a relatively small B-C reinforcement produces a statistically detectable improvement in the graph metric.

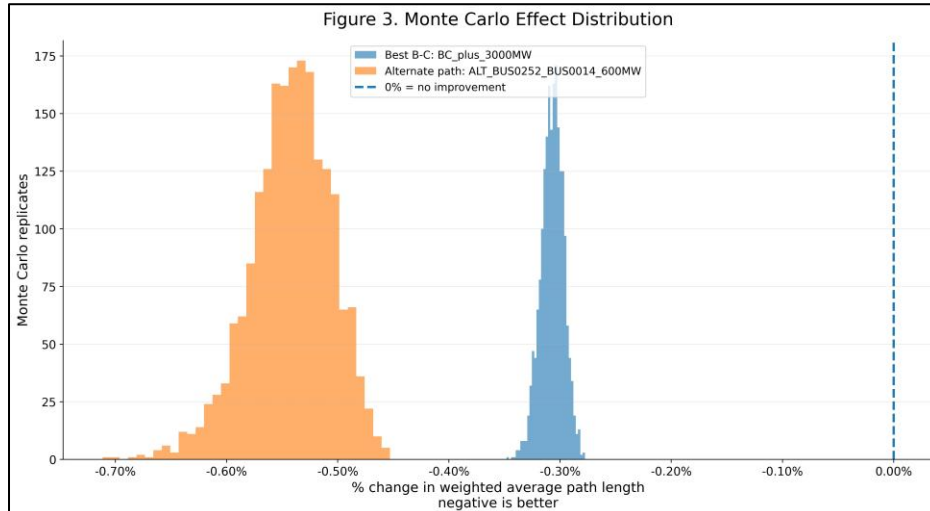
The best B-C reinforcement by mean APL improvement is BC_plus_3000MW. It has a mean absolute APL change of -0.000112 and a mean percent change of -0.307%. Its 95% CI, [-0.328%, -0.288%], stays below zero, so the intervention is statistically beneficial. Figure 2 pictorially shows full sweep:



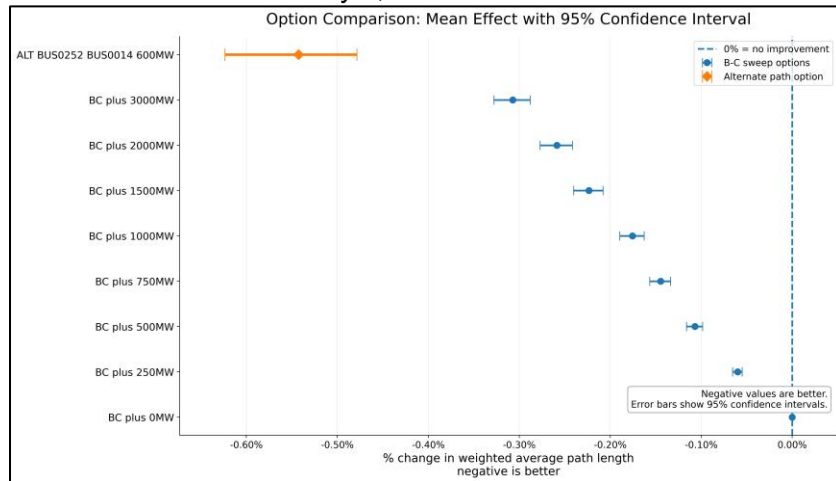
Result 2: the tested alternate path beats the best B-C reinforcement

The alternate path ALT_BUS0252_BUS0014_600MW has a mean APL change of -0.000197, with a 95% CI of [-0.000226, -0.000173]. In percent terms, it improves weighted average path length by -0.543% with a 95% CI of [-0.623%, -0.478%].

Compared with the best B-C reinforcement, the alternate path produces about 1.77 times the mean percent improvement. In this model, redundancy out of Region C provides more graph-level benefit than adding more capacity to the same B-C bottleneck. The alternate path is statistically significant, but the effect size remains modest, so it should be treated as a screening recommendation rather than a construction decision.



Alternate path is best by means of improvement proposed.



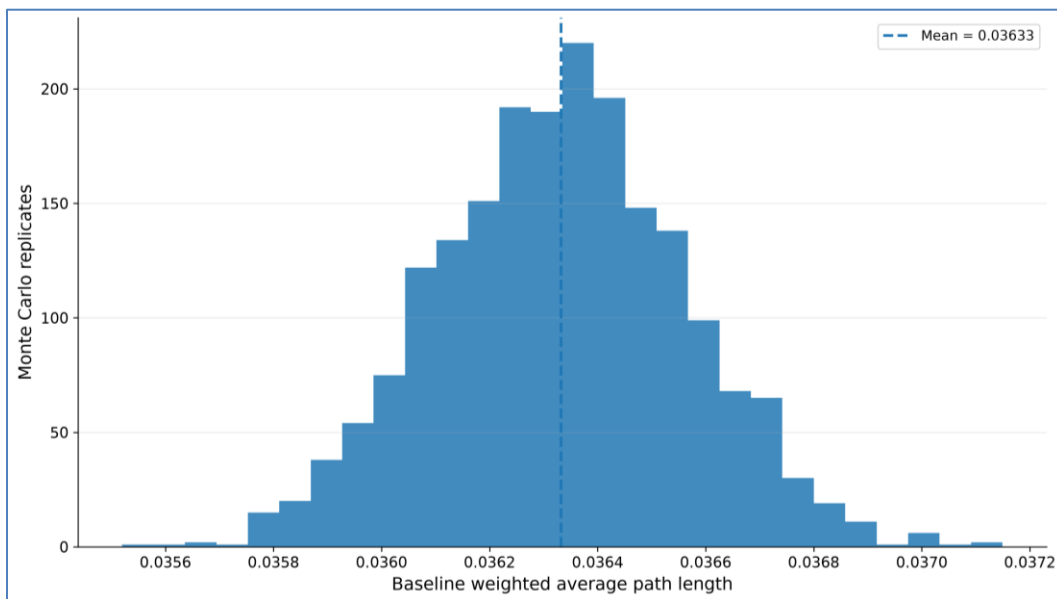
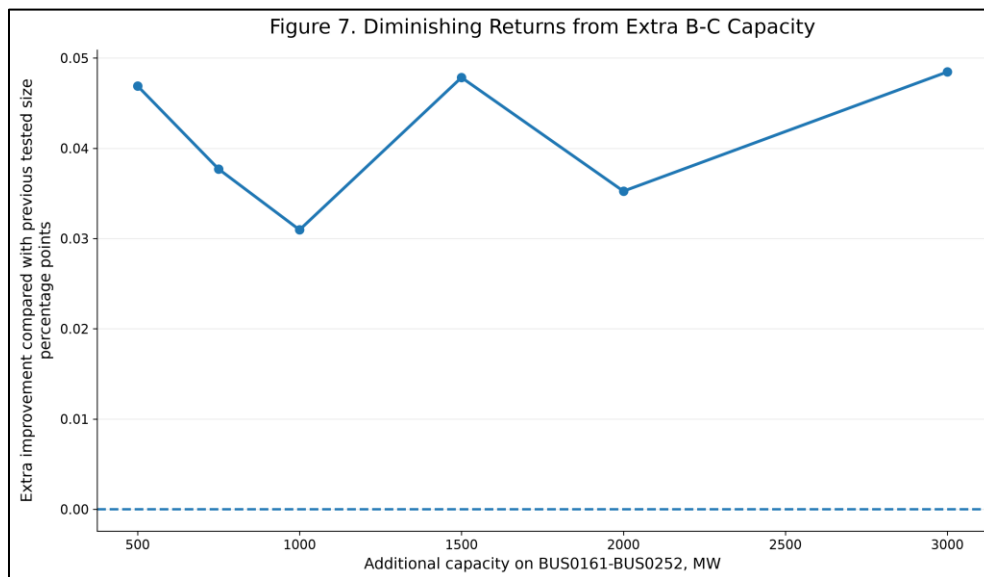
What Really Matters

The alternate-path option is the stronger intervention in this graph model. In the $R = 2,000$ Monte Carlo run, ALT_BUS0252_BUS0014_600MW improves weighted average path length by -0.543%, compared with -0.307% for the best B-C reinforcement, BC_plus_3000MW. **In plain terms, the alternate path gives about 1.77× more improvement because it adds a second route out of Region C instead of only making the same bottleneck line larger.** Both, confidence intervals stay below zero, so both upgrades are statistically beneficial. The difference is that the alternate path adds a new route out of Region C, while B-C reinforcement only adds capacity to the same existing dependency.

What the Analysis Reveals — and What It Does Not

This result should be treated as a screening finding, not a construction recommendation. Weighted average path length is a structural graph metric, not a physical power-flow result. The model does not include AC power flow, voltage limits, relay behavior, siting, cost, permitting, or construction risk. The practical conclusion is therefore: prioritize the alternate-path concept for engineering review, while treating B-C reinforcement as a real but smaller graph-level improvement.

Critical interpretation: The network has 300 buses and 422 transmission lines. Region C has 20,667 MW of generation but only 254 MW of load, while Regions A+B have 20,999 MW of load. The B-C connection is only 1,800 MW, and there is 0 MW direct A-C capacity. In the $R = 2,000$ Monte Carlo run, the best B-C reinforcement improves weighted **APL by -0.307%** with a 95% CI of [-0.328%, -0.288%], while the alternate path improves it by -0.543% with a 95% CI of [-0.623%, -0.478%]. That makes the alternate path about **1.77× stronger by mean improvement.**

Additional Diagrams:**#1: Baseline weighted Average Path Length (APL) distribution:****#2: B-C extra capacity diminishing returns:**

Adding more capacity to the B-C line still helps, but each extra upgrade gives less new benefit than the one before it. In simple terms, after a point, making the same line bigger is not as useful as adding another route.

Decision Table:

The decision table shows that both options are real improvements, but the alternate path is stronger: it improves weighted average path length by **-0.543%** with a 95% CI of [-0.623%, -0.478%], compared with **-0.307%** and [-0.328%, -0.288%] for the best B-C reinforcement. In plain terms, the alternate path gives about 1.77× more improvement because it adds a second route out of Region C instead of only making the same bottleneck line larger.

Decision item	Option	Mean % change in weighted APL	95% CI	Decision meaning
Best B-C reinforcement	BC_plus_3000MW	-0.307%	[-0.328%, -0.288%]	statistically beneficial, but smaller than alternate path.
Smallest B-C upgrade with CI below zero	BC_plus_250MW	-0.060%	[-0.065%, -0.055%]	Smallest tested reinforcement is statistically real.
Alternate-path option	ALT_BUS0252_BUS0014_600MW	-0.543%	[-0.623%, -0.478%]	Best tested option by mean improvement; still below practical threshold.
Recommended decision	Prioritize alternate-path concept for study	NA	NA	Do not overclaim construction readiness; recommend engineering validation.